Edge Computing enabled Battery State of Charge(SoC) estimation using TinyML techniques

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2024 IEEE International Conference on Power Electronics, Drives and Energy Systems (PEDES) | 979-8-3503-7247-2/24/$31.00 ©2024 IEEE | DOI: 10.1109/PEDES61459.2024.10961505

***Abstract*—Redox flow batteries (RFBs) are electrolyte-based batteries that have become a research hotspot due to their modular nature and decoupled power-energy characteristics. Household and microgrid environments extensively utilize RFBs as storage systems. The nonlinear chemical nature of RFBs hinders the modeling process, estimation of critical parameters like State of Charge (SoC), and consequently, overall energy management. The current methods in the literature use computationally complex multi-physical models that are unsuitable for deployment at the edge. Therefore, this study focuses on developing a resource-optimized machine learning-based SoC estimation model fit for deployment on an ESP32-based edge device. The study also outlines and enumerates the experimental setup for two types of RFBs: vanadium-based (AVRFB) and zinc bromide-based (ZBRFB), its data preprocessing steps, and the edge deployment process. Additionally, model quantization has also been targeted to reduce computational requirements. Using a dynamically quantized model, the devised framework achieved a R^2 score of 0.975 for AVRFB, performing inference in 1.72 seconds. In the case of ZBRFB, the SOC estimation model achieved a R^2 score of 0.818 with the float32 quantized model, outputting inference in 3.18 seconds.**

***Index Terms*—Redox flow batteries, State of Charge, TinyML, Estimation**

I. Introduction

The progressive electrification and continuous integration of Renewable Energy Sources (RESs) in MicroGrids (MGs) and household settings have led to higher penetration of Redox Flow Batteries (RFBs) as battery energy storage systems (BESSs). RFBs are utilized with RESs to help store excess available energy and provide ancillary support, such as voltage and frequency regulation, as problems arise due to the RESs coupling with the traditional power system [1]. Multiple authors have proposed an energy management system (EMS) to promote cost effectiveness, reliability, and stability of the grid system and household settings, considering the advantages and ensuring proper working of the BESSs [2, 3, 4]. Kumar et al. [3] devised an energy management agent using an optimization algorithm and simulate an MG environment with BESSs and electric vehicles in order to study how BESSs affect MG operations and energy management. Baberwal et al. [2] suggested the use of reinforcement learning-based Q-learning as a decision-making agent in EMS for a residential environment featuring BESS and PV systems. The authors of [4] conducted a energy management study for an EV charging station utilizing Vanadium-based RFB (VRFBs), solar PV and local grid.

In literature, all the EMS employs State-of-Charge (SoC), an immeasurable quantity, as a decision-making parameter for BESSs. SoC quantifies the remaining available capacity of a given battery system, $Q_{curr}(t)$, to the maximum nominal capacity, Q_{max}, and can mathematically be presented as:

$$SoC = \frac{Q_{curr}(t)}{Q_{max}} * 100\% \tag{1}$$

The literature currently divides SoC estimation methodologies into three categories: conventional-based, equivalent model-based, and data driven-based techniques. Conventional-based techniques widely use the columb-counting method, which performs the integration of measured current over time from the initial SoC level. The initial SoC level is based on open-circuit voltage (OCV) measurement. The method is prone to error due to uncertain driving cycles and conditions faced during operation, making it difficult to perform OCV [5].

The equivalent model-based methods formulate electrical or electrochemical circuit models that closely adopt the dynamic behavior and operation of BESSs. For RFBs, Binyu et al. [6] devise a thermal-dependent electrical circuit model incorporating the charge/discharge characteristics of VRFBs. Using the Extended Kalman filter technique, the model SOC estimation error is below 0.05. In [7], a dynamic multi-cell

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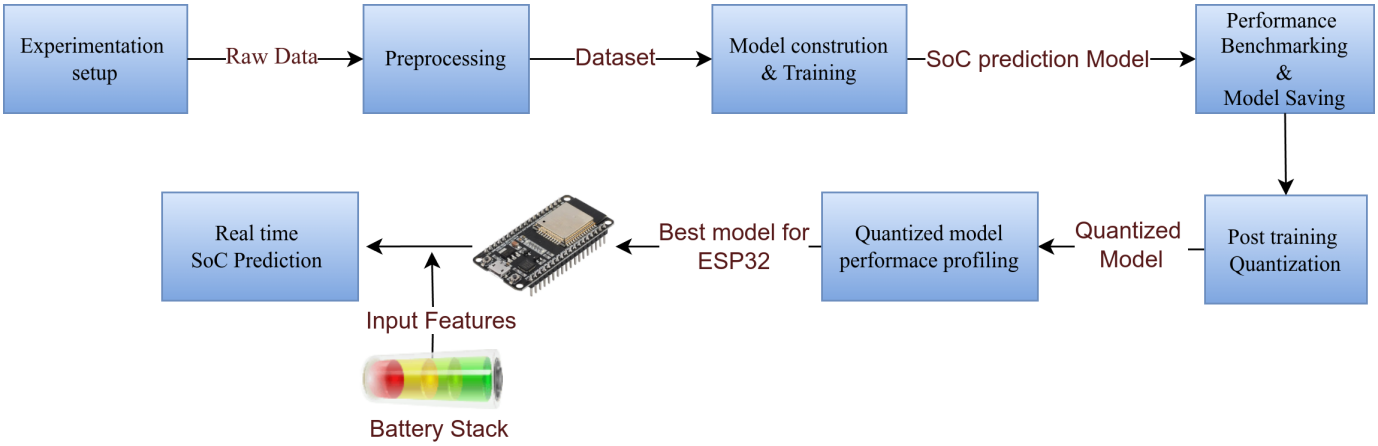

Fig. 1: The implemented pipeline

equivalent model as a lumped model is designed for 40-cell stacks for VRFBs. The authors validated the formulated model against electrochemical impedance spectroscopy measurement and found it to be as accurate.

The model-based methods are highly accurate, but equivalent model parameters are difficult to obtain for RFBs due to their chemical and hydraulic components. The methodology also has its drawbacks for real-time implementation due to model complexity [8]. Therefore, researchers are choosing data-driven methods for SoC estimation instead of conventional and model-based approaches. The data-driven approach utilizes machine learning (ML) and deep learning (DL) algorithms to build a relationship between the available feature set and the target. Niu et al. [9] used a backpropagation neural network as VRFBs SoC estimator. The SoC estimator is optimized using Levenberg-Marquardt and Bayesian regulation optimization approaches. In a similar study, a backpropagation neural network is used for SoC estimation, using loss degree capacity prediction from a probabilistic neural network as an input feature [10]. In a different study, Yadav et al. [11] compared various ML prediction algorithms for estimating the SoC for two types of RFBs. The long-short-term memory model achieved a lowest root mean square error score of 2.010e-03.

The utilization of data-driven SoC estimators also offers functionality to handle intricate dataset characteristics, is scalable, and is easily implemented. However, sophisticated data-driven models require high computational power and cannot be deployed in resource-constrained environments. To overcome this issue, researchers from academia and industry have suggested using a cloud computing paradigm to meet computational requirements [12]. Wu et al. [13] implemented a random forest regression model on a cloud to estimate degradation in BESSs. An edge device transfers the input feature to the cloud after extracting it using the 3-round feature selection approach. In a similar study, Shi et al. [14] used a similar edge-cloud computing system to estimate SoC and degradation using transformer-based neural networks. However, this approach requires a high-bandwidth internet throughout the application's lifetime, harbors data breach concerns, and has ongoing operational costs. Therefore, it is not preferred in constrained environments. This study suggests a step-by-step process for estimating SoCs at the edge using the tinyML framework and a microcontroller for two different RFBs. The contributions of the proposed work are:

- This paper presents an adaptable experimental setup to assist the collection of data for two specific types of (RFBs): the All-Vanadium RFB (AVRFB) and the Zinc-Bromine RFB (ZBRFB).
- The paper also outlines data preprocessing, model training, and quantization steps to estimate SoC at Edge.
- Quantitative comparison, including quantization profiling and inference analysis of the different quantized models, are presented to determine the best deployment model.

The rest of the article is organized as: Section II contains information on the experimental design, feature engineering, and the model used for SoC estimation. The detailed benchmarking result of the employed model, quantized model, and inference analysis is provided in Section III. The study's conclusion is presented in Section IV.

II. Methodology

The following subsections describe the individual components of this study. Firstly, an experimental setup is designed for AVRFB and ZBRFB and is used for data recording. The collected raw data is preprocessed and forwarded to the ML model for training. The trained model is benchmarked and saved for post-training quantization. As shown in Fig. 1, the saved model is then quantized, and its performance is analyzed to find the best quantized strategy for use on the ESP32 microcontroller. The ESP32 microcontroller is deployed for SoC prediction.

A. Experimental design

1) All Vanadium RFB: The devised AVRFB stack consists of 21 cells, each developed using a 6 mm graphite plate with a flow pattern. Each cell uses a Nafion-117 membrane

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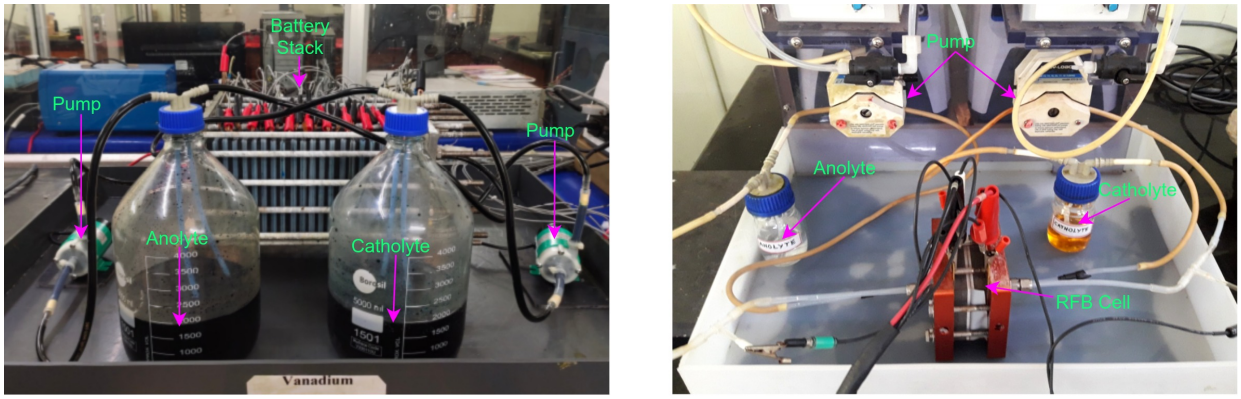

a) AVRFB stack

b) ZBRFB cell

Fig. 2: Battery experimental setup [11]

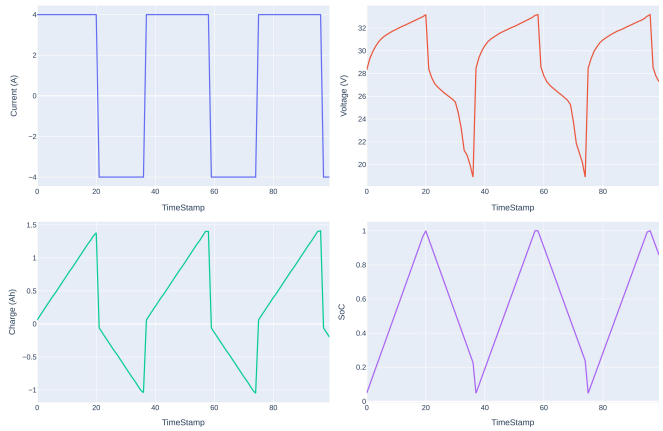

Fig. 3: Preprocessed data for AVRFB

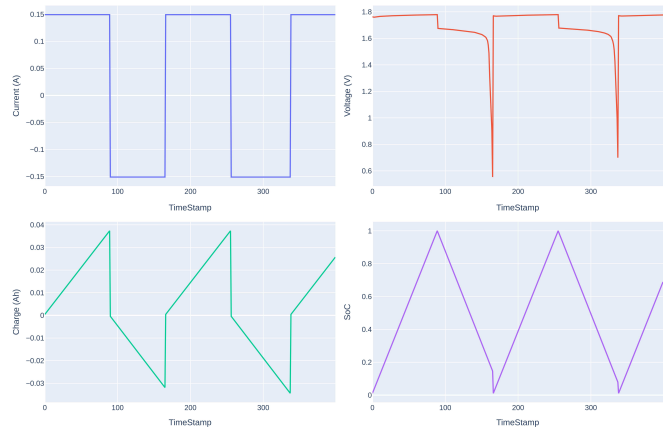

Fig. 4: Preprocessed data for zinc bromine redox flow cell

to help ions move between the electrolytes in the reaction chamber. The electrolyte was made up of two 2.5-liter parts: the anolyte and the catholyte solutions. Both of them had the same amount of 2 Mole (M) vanadium ions and 4.5 M sulfuric acid. The Bitrode battery tester system (FTV2-100-100, USA) is employed for galvanic charge-discharge analysis (GCDA) for 100 cycles with a flow rate of 2 liters per minute,. Fig. 2(a) showcases the experimental setup of the AVRFB stack.

2) Zinc-Bromine RFB: The ZBRFB cell comprises two graphite plates with a 6 mm grooved section on each plate. The single cell uses a Daramic polyethylene electrolyte separator to facilitate ionic exchange. This electrolyte was made up of 50 mL of each anolyte and catholyte solution having an identical composition of 3 M $ZnBr^2$ + 3 M KCl + 1:1 ratio of (0.89 M) $N-Methyl-N-EthylMorpholiniumbromide$ and (0.96 M) $N-Methyl-N-EthylPyrrolidiniumBromide$. The Scribner redox flow cell test system (Model: 857, USA) is utilized to conduct the experiment and collect data for 300 cycles. The cell's flow rate is set to 30 mL/minute to conduct GCDA. Fig. 2(b) showcases the experimental setup of the ZBRFB cell.

B. Data collection and preprocessing

In the case of AVRFB, the Bitrode battery tester records a total of 11 feature sets for 100 cycles at 1-minute intervals. To increase the number of available datasets, each cell in the stack was treated as a stand-alone battery system, resulting in just over 1,10,000 data points. For ZBRFB, the Scribner battery tester recorded 16 features set with 10-second intervals for 300 charging and discharging cycles, resulting in just over 52,000 data points. For both battery type, current (I), voltage (V), and charge (Ah) are selected as input features as determined in a prior study [11]. In the previous investigation, Hyperopt Optimizer [15] was used to choose these feature sets from all available options.

For both datasets, feature normalization was performed to standardize all the input features to a range of [0,1] to facilitate better learning. As SoC is an immeasurable parameter, ground truth to train the SoC prediction is unavailable. Therefore, ground truth is generated using the coulomb counting technique. Coulomb counting keeps track of SoC in real time by integrating the battery current over time and by adding previous SoC to it, as shown in Equation 2.

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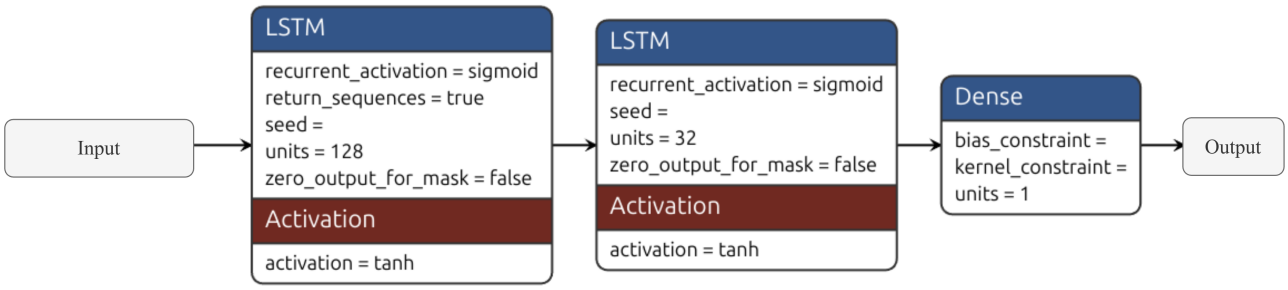

Fig. 5: SoC estimator architecture

$$SoC_{t+1} = SoC_t + \frac{1}{NC}\int_{t_0}^{t_0+d\tau} I(\tau)d\tau \tag{2}$$

where NC is BESSs nominal capacity, $I(\tau)$ is the charging/discharging current value, and SoC_t is previous SoC value at timestamp t. The resultant features and target for each battery type are graphically presented in Fig. 3 and 4.

C. SoC estimator architecture

In this study, the Long Short-Term Memory (LSTM) architecture [16] is used to build a deep learning-based SoC estimator. The designed SoC estimator consists of 3 layers, excluding the input layer. As shown in Fig. 5, there are 2 layers belonging to the LSTM network and having cell numbers of 128 and 32, respectively. The output layer is a fully connected dense layer with node size 1. The estimator employs an Adam optimizer with a learning rate of 0.001 and incorporates a stopping criteria with a tolerance of 0.001 with a threshold value at 10 epochs.

The selection of LSTM-based SoC estimator is based on a previous study [11] in which the LSTM-based model presented comparable performance to transformer-based models and outperformed the recurrent neural network model. The transformer-based model, such as the Temporal Fusion Transformer [17], is computationally expensive and not feasible to deploy on low-computing edge devices. A single LSTM cell consists of three gates: a forget gate, an input gate, and an output gate, which facilitate the acquisition of long-term dependencies. The forget gate utilizes a sigmoid function to determine the information to be retained, while the cell state maintains the information transmitted from previous layers to the current layer.

D. Considered hardware

This study used the ESP32 microcontroller as an edge device. The ESP32 is renowned for its low cost and exceptional energy efficiency, making it the best choice for deployment in a constrained environment. The ESP32 integrates Wi-Fi and Bluetooth connectivity with a robust processor with 520 KB of SRAM. The SRAM is divided into two parts: IRAM, which is around 200 KB and is utilized for storing program instructions, and DRAM, which is approximately 320 KB. The input feature measuring sensors can be set up to connect directly to the ESP32 module through pins, or they can use Wi-Fi and Bluetooth to send data points wirelessly and show the results on a screen or webpage. In this work, parameters used as input to estimate SoC were transmitted via pins.

E. Quantization overview

Quantization techniques are employed to reduce the overall computational and memory footprints associated with real-time inference. The quantization process converts the value of weights and the range of the activation function from a high-precision data type to low-precision data types, such as 8-bit integers. Using low-precision data types makes the model smaller, needs less memory, and can do operations like matrix multiplication faster, enabling the deployment of ML models on low-cost microcontrollers.

In literature, there are two types of quantization techniques that exist: a) Post-training quantization; b) Quantization-aware training. The post-training quantization technique allows to reduce the computational and memory footprints of pre-trained models. On the other hand, quantization-aware training involves incorporating quantization operations within the training code of the build model and the quantized model is fine-tuned during training. In this paper, the post-training quantization technique is employed.

III. Results and Discussion

The SoC estimator was trained and validated using a single GeForce RTX 3060 GPU system with 12 GB of graphical memory for 100 epochs. Mean squared error (MSE), root mean square error (RMSE), and mean absolute error (MAE) are used to assess the accuracy of the estimator. The dataset was bifurcated into training, validation, and testing datasets using a ratio of 0.8:0.1:0.1.

A. Prediction accuracy

The performance of the LSTM model on used metrics for AVRFB and ZBRFB batteries is shown in Table I. For AVRFB, the LSTM model scored an RMSE, MAE, and MSE metrics score of 0.0103, 0.0048, and 1.058e-04 for the training dataset while having similar performance on the validation set, proving the SOC estimation model is unbiased. In validation set for AVRFB, the LSTM model attained 0.0920, 0.0028, and 8.5107e-05 as RMSE, MAE, and MSE metrics scores

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TABLE I: Model performance for SoC estimation

Battery type	Model parameters	Training			Validation		
		RMSE	MAE	MSE	RMSE	MAE	MSE
AVRFB	88737	0.0103	0.0048	1.0586e-04	0.0920	0.0028	8.5107e-05
ZBRFB	88737	0.0043	0.0030	1.8633e-05	0.0027	0.0013	7.3301e-05

Similarly, the LSTM model achieved even better RMSE, MAE, and MSE metrics scores of 0.0043, 0.0030, and 1.86e-05 on the training dataset for ZBRFB. The RMSE, MAE, and MSE metrics scores for the validation set are 0.0027, 0.0013, and 7.33e-05. As the number of features was similar for both the battery sets, the number of model parameters in both cases is constant and is 88737.

B. Quantization profiling

The results of the different precision data types on the developed model size and performance for AVRFB and ZBRFB, utilizing the testing dataset, are presented in the table. II and III. Float 16 results are unavailable due to an inbuilt error of the TensorFlow library for LSTM cells. The dynamic quantized model uses a representative dataset during the quantization process to estimate the range of all floating-point tensors. In this study, 300 random samples of data are provided as a representative dataset.

TABLE II: Quantized model performance for AVRFB

Post-training quantization	**Model Size [KB]**	**RMSE**	**MAE**	**MSE**	R^2 **score**
Keras model	1100	0.0195	0.0124	3.8176e-04	0.995
Float 32	360.6	0.056	0.043	0.003	0.960
Float 16	-	-	-	-	-
Int 8	103.0	0.434	0.545	0.373	-3.894
Dynamic	97.8	0.043	0.035	0.002	0.975

TABLE III: Quantized model performance for ZBRFB

Post-training quantization	**Model Size [KB]**	**RMSE**	**MAE**	**MSE**	R^2 **score**
Keras model	1100	0.0056	0.0044	3.1149e-05	1.00
Float 32	360.6	0.121	0.104	0.015	0.818
Float 16	-	-	-	-	-
Int 8	103.0	0.417	0.482	0.313	-2.870
Dynamic	97.8	0.355	0.233	0.062	0.229

For AVRFB, Dynamic and Float 32 achieved competitive R^2 score, similar to Keras model, with 91% and 67% size reductions, respectively. The other metrices, RMSE and MAE, witnessed some degradation as expected to low precision calculation. Contrarily, the quantized model for ZBRFB achieved similar size reduction of 91% but had high degradation in performance, achieving poor R^2 score. Float 16 quantized model performance for ZBRFB were in acceptable range and much higher compared to dynamic model. Int 8 quantized performed poorly for both battery types, scoring a negative R^2 score.

C. Inference analysis

The inference and hardware performance results of the float32 and dynamic quantized model on the ESP32 microcontroller for AVRFB and ZBRFB are presented in the table. IV and V. The dynamic quantized model is consuming less RAM and ROM of the ESP32 microcontroller and has a lower inference time. It is taking 1.72 seconds to output the SoC reading compared to 2.76 by the Float32 model for AVRFB. For ZBRFB, the dynamic quantized model is taking 2.54s to produce an inference compared to 3.18s by the Float 32 model.

TABLE IV: ESP32 device performance for AVRFB

Post-training quantization	**RAM [KB]**	**ROM[KB]**	**Inference [S]**
Float 32	15.71	401.7	2.796
Dynamic	9.50	142.03	1.72

TABLE V: ESP32 device performance for ZBRFB

Post-training quantization	**RAM [KB]**	**ROM[KB]**	**Inference [S]**
Float 32	15.71	401.7	3.18
Dynamic	9.5	142.05	2.548

IV. Conclusion

This study presents a deep learning-based state-of-charge estimation framework for 2 types of redox flow batteries, deployed using an ESP32 microcontroller. According to the results of the predictions, the employed LSTM model achieves an impressive RMSE score of 0.0103 and 0.092 on training and validation for AVRFB type RFB battery and achieves an RMSE score of 0.0043 and 0.0027 for ZBRFB. The computational and memory cost of the pre-trained model were reduce using post-training quantization by reducing model size and float-point precision requirement to achieve better inference time. According to the results, the dynamic quantize for AVRFB and float 32 for ZBRFB had competitive performance to the high precision model. The result obtained in inference analysis demonstrated that a dynamically quantized model uses less RAM and ROM and has a lower inference time.

Potential future research includes addressing battery deterioration in RFB systems and the integration of the developed mechanism with an energy management system.

Acknowledgment

The authors express their gratitude to the Director of CSIR-CEERI, Pilani, and the Director of CSIR-CECRI, Karaikudi, India, for assistance in carrying out this research. The authors would like to acknowledge the Council of Scientific and

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Industrial Research (CSIR), New Delhi, India, for sanctioning this project (HCP:0053) under Grant No: 31-2(281)/2023-2024.

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